

Indian Statistical Institute

BSDS IInd Year

Academic Year 2025 - 2026: Semester I

Course: Probability II

Instructor: Antar Bandyopadhyay

Midterm Examination

Date: October 07, 2025

Time: 11:00 AM - 01:00 PM

Total Points: 30

Duration: 120 minutes

Note: There are four problems with a total of 36 points. Solve as many as you can. Maximum you can score is 30. Show all your works. Use only the pages provided. The rough works must be done on the pages provided for the same. Do not forget to write your name and roll number. Also circle the location/centre you are attending. Good luck!

Name: _____

Roll No.: _____

Location/Centre: Kolkata Delhi Bangalore

Please DO NOT write any thing below
For Instructor/TAI use only

Problems	1	2	3	4	Total
Points					

[Please Only Turn at 11:00 AM]

[Please DO NOT write anything on this page!]

The problems are:

1. Suppose X_1 and X_2 are normally distributed with means $\mu_1, \mu_2 \in \mathbb{R}$ and variances $\sigma_1^2, \sigma_2^2 > 0$ respectively. Find necessary and sufficient condition on the parameters so that $(X_1 + X_2)$ and $(X_1 - X_2)$ are independent. [9]

2. Suppose $U \sim \text{Uniform}[0, 1]$.

(a) Find $\mathbf{E}[U^n]$, $n \geq 1$. [3]

(b) Show that for any $\varepsilon > 0$, $\mathbf{P}(U^n > \varepsilon) \rightarrow 0$ as $n \rightarrow \infty$. [6]

3. Suppose $X_1, X_2, \dots, X_n, \dots$ be *i.i.d.* random variables with Cauchy distribution. Then show that

$$\frac{X_n}{n^\alpha} \rightarrow 0 \text{ a.s.}$$

whenever $\alpha > 1$. [9]

4. $\{\varepsilon_i\}_{i \geq 0}$ are *i.i.d.* standard normal random variables. For $a \in \mathbb{R}$, define

$$X_0 = \varepsilon_0; \text{ and}$$

$$X_k = aX_{k-1} + \varepsilon_k, \text{ for } k \geq 1.$$

For $n \geq 2$, find the joint distribution of the random vector $(X_1, X_2, \dots, X_n)$.

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Rough Work Page

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